

DNS Practical Recreation Instruction Manual

SIT716 HD Plus Project - DNS normal resolution, rogue DNS response testing, and Wireshark evidence workflow

Purpose: This manual explains how another user can import the provided Ubuntu OVA, open the MiniEdit/Mininet lab, reproduce normal DNS resolution and rogue DNS response behaviour, capture packets in Wireshark, and compare the results with the website simulator.

Important: Use this manual together with the supplied OVA, DNS PCAP files, and website DNS page. If the OVA already contains prepared scripts or topology files, use those prepared files first and use the commands below as the recreation reference.

1. Required Files and Software

Download these artefacts from the DNS project folder before starting:

Item	Purpose
DNS OVA file	Ubuntu VM containing or supporting the DNS practical lab environment
DNS PCAP files	Evidence files that can be opened directly in Wireshark
Website project folder	Interactive simulator used for comparing visual output with packet evidence
This manual	Step-by-step guide for recreation and verification

Install or confirm these applications on the host computer: Oracle VirtualBox, a modern web browser, and a PDF reader. Inside the VM, confirm Ubuntu, MiniEdit/Mininet, terminal, and Wireshark are available.

2. Import the Ubuntu OVA into VirtualBox

1. Open Oracle VirtualBox. 2. Select **File -> Import Appliance**. 3. Browse to the downloaded DNS OVA file. 4. Keep the default import settings unless your machine has limited RAM or CPU. 5. Click **Finish** or **Import**. 6. Start the imported Ubuntu VM.

After the VM starts, confirm that the desktop loads correctly and open a terminal.

```
whoami
pwd
ip addr
which wireshark || wireshark --version
which mn || mn --version
```

Expected result: Ubuntu should open successfully, terminal commands should work, and Mininet/Wireshark should be available or already installed in the VM.

3. Prepare the DNS Lab Environment

Clean any previous Mininet state before starting a new run. This reduces errors caused by stale interfaces or old Mininet processes.

```
sudo mn -c
```

Open MiniEdit/Mininet. The exact path may differ depending on the OVA, but common commands are:

```
sudo python3 /usr/lib/python3/dist-packages/mininet/examples/miniedit.py
# or
sudo python3 /usr/share/mininet/examples/miniedit.py
# or find it first
sudo find /usr -iname "miniedit.py" 2>/dev/null
```

Open the prepared DNS topology if one is provided in the OVA. Start the topology from MiniEdit. Then open terminals for the client, legitimate DNS server, and attacker or rogue DNS host.

```
# Common Mininet CLI checks
nodes
net
dump
xterm h1 h2 h3
```

4. Suggested Host Role and IP Reference

Use the OVA-provided topology and notes if they define different host names. The following mapping matches the DNS evidence workflow used for the project screenshots.

Role	Example host	Example IP	Purpose
Client	h1	192.168.70.100	Sends DNS queries
Legitimate DNS server	h2	192.168.70.2	Returns correct DNS answer
Rogue DNS attacker	h3	192.168.70.50	Returns forged DNS answer
Example legitimate answer	-	192.168.70.10	Correct service address
Example rogue answer	-	10.10.10.66	Fake or malicious service address

If your VM uses different IP addresses, keep the logic the same: client queries a legitimate DNS server for normal testing, then queries or receives an answer from the rogue DNS responder for attack testing.

5. Run Normal DNS Resolution Test

5.1 Start Wireshark Capture

Open Wireshark and capture on the switch or relevant interface. In the display filter box, use:

```
dns or udp.port == 53 or arp
```

This filter shows DNS query-response traffic and ARP frames that may appear before DNS communication.

5.2 Start a Legitimate DNS Service

If the OVA includes a prepared DNS script, run that script. Otherwise, this dnsmasq example can be used inside the legitimate DNS server host terminal. Adjust the interface/IP if needed.

```
sudo dnsmasq --no-daemon --log-queries --no-resolv --listen-address=192.168.70.2
--bind-interfaces --address=/example.local/192.168.70.10
```

5.3 Send a DNS Query from the Client

From the client host, query the legitimate DNS server. Use dig if available, otherwise nslookup.

```
dig @192.168.70.2 example.local
# alternative
nslookup example.local 192.168.70.2
```

Expected Wireshark result: a standard DNS query from the client to the legitimate DNS server, followed by a standard DNS response from the legitimate DNS server to the client.

6. Run Rogue DNS Response Test

Stop or leave the legitimate test running depending on your demonstration design. The attack test should make clear that a rogue responder can return a forged address.

6.1 Start Rogue DNS Service on the Attacker

```
sudo dnsmasq --no-daemon --log-queries --no-resolv --listen-address=192.168.70.50
--bind-interfaces --address=/example.local/10.10.10.66
```

6.2 Query the Rogue DNS Responder

From the client, query the rogue DNS server. This represents a rogue DNS response or malicious resolver scenario.

```
dig @192.168.70.50 example.local
# alternative
nslookup example.local 192.168.70.50
```

Expected Wireshark result: a DNS query involving the rogue responder and a response that returns the forged IP address. Compare the DNS answer field with the legitimate result from the previous section.

7. Save Evidence and PCAP Files

In Wireshark, save the DNS capture after each run or save one combined capture with clear packet numbers. Use descriptive filenames so the evidence can be reused in the website and report.

```
dns-normal-query-response.pcapng
dns-rogue-query-response.pcapng
```

Recommended screenshots: normal DNS query packet details, normal DNS response packet details, rogue DNS query-response packet details, and a final view showing the filter used in Wireshark.

8. Troubleshooting Checklist

Problem	Fix
No DNS packets visible	Confirm the Wireshark interface is correct and use filter: dns or udp.port == 53
dig command missing	Install dnsutils or use nslookup if available
dnsmasq already running	Stop old dnsmasq process or run sudo mn -c and restart the lab
No response from DNS server	Check IP addresses, host connectivity with ping, and listen-address in dnsmasq
Wrong answer returned	Check which DNS server IP the client queried and confirm the address rule in dnsmasq

9. Compare with the Website Simulator

Open the website DNS page and run the predefined normal and attack simulations. The normal website flow should match the legitimate DNS query-response evidence. The attack website flow should match the rogue DNS response evidence, where the client is redirected to a forged destination address.

Use the website event log, packet animation, Wireshark packet list, and saved PCAP files together as the final evidence chain.

10. Final Verification Checklist

Before submitting or sharing the DNS practical package, confirm all items below are complete:

- ☐ OVA imports and starts successfully in VirtualBox.
- ☐ MiniEdit/Mininet topology starts without stale interface errors.
- ☐ Wireshark captures DNS packets using dns or udp.port == 53.
- ☐ Normal DNS query and response are visible in Wireshark.
- ☐ Rogue DNS response or forged DNS answer is visible in Wireshark.
- ☐ PCAP files are saved with clear filenames.
- ☐ Website DNS page links to this instruction manual and the DNS artefact folder.